FLAT BASE CHANGE FOR THE COHOMOLOGY OF A QUASI-COHERENT MODULE ON AN ALGEBRAIC SPACE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-spaces-cohomology/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Flatness of the base changed morphism, and flatness of the induced morphisms of small etale sites. Morphisms of Spaces, labels spaces-morphisms-lemma-base-change-flat (invoked at proof line 1950) and spaces-morphisms-lemma-flat-morphism-sites (invoked at proof line 1953). They show that g' is flat and that flatness of g and g' is the flatness of the corresponding morphisms of small etale ringed sites, which is what the base change map requires.
- The base change map for a flat morphism of sites. Cohomology on Sites, label sites-cohomology-lemma-base-change-map-flat-case (invoked at proof line 1956). It supplies the map $g^*R^pf_*\mathcal{F} \to R^pf'_*\mathcal{F}'$ whose bijectivity is the content of part (1).
- Vanishing of the higher direct images of the affine morphism g', with Leray. Same chapter spaces-cohomology.tex, label lemma-affine-vanishing-higher-direct-images (invoked at proof line 1977) and Cohomology on Sites, label sites-cohomology-lemma-apply-Leray (invoked at proof line 1979). They give $H^p(X_B, \mathcal{F}_B) = H^p(X, g'_*\mathcal{F}_B)$, because g' is affine as a base change of the affine morphism $\Spec(B) \to \Spec(A)$.
- Lazard's theorem, writing a flat algebra as a filtered colimit of finite free modules. Algebra, label algebra-theorem-lazard (invoked at proof line 1997). It provides the filtered system of finite free A-modules M_i with $B = \colim M_i$, over which the tensor product and the cohomology can be moved.

Date: stacks-project:spaces-cohomology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

- Cohomology of a quasi-compact quasi-separated space commutes with filtered colimits. Same chapter spaces-cohomology.tex, label lemma-colimits (invoked at proof line 2014). It is the third equality in the chain, and it is what allows the colimit to be moved out of the cohomology of the tensor product.

Source reference. Original theorem 058G: sources/stacks-chow/tags.txt, source lines 6779–6779. Navigation pointer for \ref{algebra-theorem-lazard} used inside the counted span (line 1997); the argument that consumes it is retained in the span.

Source reference. Original lemma 073H: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 1522–1522. Navigation pointer for \ref{lemma-affine-vanishing-higher-direct-images} used inside the counted span (line 1977); the argument that consumes it is retained in the span.

Source reference. Original lemma 073E: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 667–667. Navigation pointer for \ref{lemma-colimits} used inside the counted span (line 2014); the argument that consumes it is retained in the span.

Source reference. Original lemma 0733: sources/stacks-chow/tags.txt, source lines 9047–9047. Navigation pointer for \ref{sites-cohomology-lemma-apply-Leray} used inside the counted span (line 1979); the argument that consumes it is retained in the span.

Source reference. Original lemma 0736: sources/stacks-chow/tags.txt, source lines 9050–9050. Navigation pointer for \ref{sites-cohomology-lemma-base-change-map-flat-case} used inside the counted span (line 1938); the argument that consumes it is retained in the span.

Source reference. Original lemma 03MO: sources/stacks-chow/tags.txt, source lines 4756–4756. Navigation pointer for \ref{spaces-morphisms-lemma-base-change-flat} used inside the counted span (line 1950); the argument that consumes it is retained in the span.

Source reference. Original lemma 073C: sources/stacks-chow/tags.txt, source lines 9056–9056. Navigation pointer for \ref{spaces-morphisms-lemma-flat-morphism-sites} used inside the counted span (line 1953); the argument that consumes it is retained in the span.

Lemma 0.1 (Flat base change)**.** *Let S be a scheme. Consider a cartesian diagram of algebraic spaces*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{g} & Y \end{array}$$

over S. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module with pullback $\mathcal{F}' = (g')^*\mathcal{F}$. Assume that g is flat and that f is quasi-compact and quasi-separated. For any $i \geq 0$

(1) *the base change map of Cohomology on Sites, Lemma 0736 is an isomorphism*
$$g^* R^i f_* \mathcal{F} \longrightarrow R^i f'_* \mathcal{F}',$$
(2) *if $Y = \mathrm{Spec}(A)$ and $Y' = \mathrm{Spec}(B)$, then $H^i(X, \mathcal{F}) \otimes_A B = H^i(X', \mathcal{F}')$.*

Proof. The morphism g' is flat by Morphisms of Spaces, Lemma 03MO. Note that flatness of g and g' is equivalent to flatness of the morphisms of small étale ringed

sites, see Morphisms of Spaces, Lemma 073C. Hence we can apply Cohomology on Sites, Lemma 0736 to obtain a base change map

$$g^* R^p f_* \mathcal{F} \longrightarrow R^p f'_* \mathcal{F}'$$

To prove this map is an isomorphism we can work locally in the étale topology on Y'. Thus we may assume that Y and Y' are affine schemes. Say $Y = \mathrm{Spec}(A)$ and $Y' = \mathrm{Spec}(B)$. In this case we are really trying to show that the map

$$H^p(X, \mathcal{F}) \otimes_A B \longrightarrow H^p(X_B, \mathcal{F}_B)$$

is an isomorphism where $X_B = \mathrm{Spec}(B) \times_{\mathrm{Spec}(A)} X$ and $\mathcal{F}_B$ is the pullback of $\mathcal{F}$ to X_B. In other words, it suffices to prove (2).

Fix $A \to B$ a flat ring map and let X be a quasi-compact and quasi-separated algebraic space over A. Note that $g' : X_B \to X$ is affine as a base change of $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$. Hence the higher direct images $R^i(g')_* \mathcal{F}_B$ are zero by Lemma 073H. Thus $H^p(X_B, \mathcal{F}_B) = H^p(X, g'_* \mathcal{F}_B)$, see Cohomology on Sites, Lemma 0733. Moreover, we have

$$g'_* \mathcal{F}_B = \mathcal{F} \otimes_{\underline{A}} \underline{B}$$

where $\underline{A}$, $\underline{B}$ denotes the constant sheaf of rings with value A, B. Namely, it is clear that there is a map from right to left. For any affine scheme U étale over X we have

$$\begin{aligned} g'_* \mathcal{F}_B(U) &= \mathcal{F}_B(\mathrm{Spec}(B) \times_{\mathrm{Spec}(A)} U) \\ &= \Gamma(\mathrm{Spec}(B) \times_{\mathrm{Spec}(A)} U, (\mathrm{Spec}(B) \times_{\mathrm{Spec}(A)} U \to U)^* \mathcal{F}|_U) \\ &= B \otimes_A \mathcal{F}(U) \end{aligned}$$

hence the map is an isomorphism. Write $B = \mathrm{colim}\, M_i$ as a filtered colimit of finite free A-modules M_i using Lazard's theorem, see Algebra, Theorem 058G. We deduce that

$$\begin{aligned} H^p(X, g'_* \mathcal{F}_B) &= H^p(X, \mathcal{F} \otimes_{\underline{A}} \underline{B}) \\ &= H^p(X, \mathrm{colim}_i\, \mathcal{F} \otimes_{\underline{A}} \underline{M_i}) \\ &= \mathrm{colim}_i\, H^p(X, \mathcal{F} \otimes_{\underline{A}} \underline{M_i}) \\ &= \mathrm{colim}_i\, H^p(X, \mathcal{F}) \otimes_A M_i \\ &= H^p(X, \mathcal{F}) \otimes_A \mathrm{colim}_i\, M_i \\ &= H^p(X, \mathcal{F}) \otimes_A B \end{aligned}$$

The first equality because $g'_* \mathcal{F}_B = \mathcal{F} \otimes_{\underline{A}} \underline{B}$ as seen above. The second because $\otimes$ commutes with colimits. The third equality because cohomology on X commutes with colimits (see Lemma 073E). The fourth equality because M_i is finite free (i.e., because cohomology commutes with finite direct sums). The fifth because $\otimes$ commutes with colimits. The sixth by choice of our system. □